

Sachin Kumar

Software Engineer II | iOS & macOS

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KEY ACHIEVEMENTS

- Cut **concurrency-related crash rates by 40%** across a fleet of **10,000+ managed macOS endpoints** by migrating legacy GCD code to Swift Concurrency (async/await, actors, structured tasks).
- Reduced **data lookup latency by 50%** and **file-transfer failure rates by 35%** by redesigning the Core Data persistence layer and rebuilding the download pipeline with resumable chunked transfers and bounded concurrent queues.
- Shipped **CodeForge** to the **App Store** as sole developer architecture, tests, submission, and App Review a native macOS and iOS IDE with a dual-path code execution engine.

SUMMARY

Software engineer with 3+ years building and operating production iOS and macOS systems at enterprise scale. Owns features end to end system design, implementation, unit and UI testing, CI/CD, code signing, and release on a platform running continuously across 10,000+ devices. Strong fundamentals in data structures, algorithms, and distributed system design, with a published App Store app and production experience in Apple's system-level frameworks (NetworkExtension, XPC, LaunchDaemon).

EXPERIENCE

42Gears Mobility Systems Pvt. Ltd.

February 2023 – Present

Bengaluru, India | Unified Endpoint Management (UEM) SaaS platform

Software Engineer, iOS & macOS

July 2023 – Present

- Cut **concurrency crash rates 40%**, measured over release-to-release crash telemetry, by migrating legacy GCD and DispatchQueue code to **Swift Concurrency** with actor-isolated shared state.
- Reduced **data lookup latency 50%** at fleet scale by redesigning the **Core Data** schema, adding indexes on hot query paths, and replacing per-row writes with batch fetch and update operations.
- Reduced **transfer failure rates 35%** by designing a resumable **chunked download system** over URLSession with progress callbacks, retry semantics, and a bounded concurrent worker pool.
- Designed and shipped privileged macOS system agents **LaunchDaemon** daemonization, **XPC** inter-process communication, and privileged helper tools running as root across **10,000+ enterprise endpoints**.
- Architected **NetworkExtension** providers (NEFilterProvider, NEAppProxyProvider) for real-time network traffic filtering and per-application data usage attribution at the OS flow layer.
- Migrated core modules from **MVVM** to **VIPER**, applying dependency-injection boundaries to make previously untestable business logic unit-testable.
- Designed a structured **analytics and event-processing pipeline** capturing device events, health metrics, and network diagnostics, serving administrator-facing reporting across the fleet.
- Owned release engineering end to end: **Playwright** and **Appium** regression automation, static analysis gates, code signing, notarization, packaging, and automated distribution through **GitLab CI/CD**.
- Eliminated a class of production security risk by migrating plaintext local storage to **CryptoKit** and **Keychain** encrypted formats and replacing deprecated third-party libraries with first-party SDKs, with zero regressions for devices on older macOS releases.
- Built an internal diagnostics platform adopted by **20+ engineers and QA** real-time log parsing, job payload inspection, and live process state inspection after identifying debugging turnaround as the team's largest cycle-time cost.
- Automated release validation with local **LLM** evaluation and **multi-agent workflows** covering build verification and deployment checks.

Software Engineer Intern, iOS

February 2023 – June 2023

- Delivered production features in **Swift**, **UIKit**, and **SwiftUI**; implemented layered local persistence across **Core Data**, **UserDefaults**, and **Keychain**; converted to full-time after 5 months.

PROJECTS

CodeForge DSA & Interview Prep IDE Swift, SwiftUI, Swift Concurrency, JavaScriptCore, XCTest Live on the App Store

- Designed a dual-path code execution engine behind a single protocol abstraction subprocess execution on macOS, JavaScriptCore on iOS so the same test-runner core works on both platforms under different sandbox constraints.
- Shipped solo to the App Store: architecture, offline-first persistence, unit tests, TestFlight beta, App Store Connect submission, and App Review approval.
- apps.apple.com/app/id6787703716 | github.com/sachin6174/SwiftForge

Agentic Gig Platform Next.js, Firebase, Python, Kotlin, Google Cloud Google Cloud Agentic AI Hackathon 2025

- Built an AI agent platform delivering agricultural advisory to farmers; selected among the top 700 teams out of 57,000+ developers. github.com/abhikumar002/Agentic-Gig-Project

TECHNICAL SKILLS

Languages: Swift, Objective-C, C++, JavaScript, Python, Bash/Shell, SQL

Systems & Concurrency: Swift Concurrency (async/await, actors, Sendable), GCD, DispatchQueue, Operation Queues, multithreading, memory management/ARC, XPC IPC, LaunchDaemon/launchd, process daemonization, NetworkExtension

Apple Frameworks: SwiftUI, UIKit, Foundation, Combine, Core Data, SwiftData, Core Location, CryptoKit, Keychain Services, Core ML, Auto Layout

CS Fundamentals: Data Structures, Algorithms, Complexity Analysis, System Design, Scalability, Load Balancing, Database Design, Object-Oriented Design, SOLID, Design Patterns

Testing & Reliability: XCTest, XCUITest, unit testing, UI testing, TDD, Xcode Instruments (Time Profiler, Leaks, Allocations), profiling, debugging, code review, SonarQube, Playwright, Appium

Infrastructure & Release: Git, GitLab CI/CD, GitHub Actions, Swift Package Manager, CocoaPods, Fastlane, TestFlight, App Store Connect, code signing, notarization, REST APIs, URLSession, JSON/Codable, SQLite, Firebase, Jira, Agile/Scrum

AI: Local LLMs, AI agents, multi-agent workflows, LLM evaluation, Claude Code, GitHub Copilot, Cursor

EDUCATION

Chandigarh Group of Colleges, Landran August 2019 – May 2023
Bachelor of Engineering in Computer Science and Engineering CGPA: 7.9/10
Mohali, Punjab, India

ADDITIONAL

- **Algorithms & fundamentals:** Certified in System Design and Data Structures in C++ (Coding Ninjas).